

Note Title

筆記標題

Your Name

July 29, 2026

Preface

Preface goes here.

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1. Introduction

Lecture 1: First Lecture

S1.1. Theorem Environments

13 Oct. 08:00

Definition 1.1.1 (Natural number). The set of **natural numbers** is denoted $\mathbb{N}$.

Lemma 1.1.2 (Useful lemma)

Given the axioms of **natural numbers** $\mathbb{N}$, we have

$$0 \neq 1.$$

An obvious proof. Obvious. ■

Proposition 1.1.3 (Useful proposition)

From **Lemma 1.1.2**, we have

$$0 < 1.$$

Theorem 1.1.4 (Mass-energy equivalence)

Given **Proposition 1.1.3**, we then have

$$E = mc^2.$$

Proof. The blank left for me is too small, hence we put the proof in **Appendix A.1**.

Proof of key step. This inner proof uses ■ to distinguish from the outer ■. ■

Corollary 1.1.5 (Riemann hypothesis)

The real part of every nontrivial zero of the Riemann zeta function is $\frac{1}{2}$.

Proof. Left to the reader. ■

DIY

S1.2. Other Environments

Exercise 1.2.1. Prove that $1 < 2$.

Solution. Iteratively apply **Proposition 1.1.3**. ■

Example 1.2.2 (Interesting example)

We note that $1 \neq 2!$

Note 1.2.1 (Important note). As a consequence, $2 \neq 3$ also.

Remark 1.2.1. See that [Proposition 1.1.3](#) is powerful. We also have a citation: [\[New26\]](#).

As previously seen. We saw [Lemma 1.1.2](#) is helpful.

Claim 1.2.1. The center $Z(G)$ is always a normal subgroup of G .

Proof. For any $z \in Z(G)$ and $g \in G$, we have $gzg^{-1} = z \in Z(G)$. ■

Intuition. Key idea: think of the center as the fixed points of conjugation.

Internal links

You can use `\autoref` for internal references and `\cite` for bibliography entries.

S1.3. Footnote Test**Theorem 1.3.1**

Every group of prime order is cyclic.^a

^aThis is a footnote inside a theorem box.

Example 1.3.2

The integers $\mathbb{Z}$ form an infinite cyclic group.^a

^aThis is a footnote inside an example box.

S1.4. Commutative Diagrams**Example 1.4.1**

A commutative square drawn with `tikz-cd`:

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & & \downarrow h \\ C & \xrightarrow{k} & D \end{array}$$

S1.5. Vocab and Dividers

Use `\vocab{term}` to highlight a **key term** in blue bold.

Use `\hrulebar` to insert a visual section divider:

Content continues after the divider.

2. Chapter Title

Lecture 2: Second Lecture

S2.1. Section Title

20 Oct. 08:00

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3. Chapter Title

Lecture 3: Third Lecture

S3.1. Section Title

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Appendix

A. ■ Additional Proofs

SA.1. Proof of Theorem 1.1.4

We can now prove Theorem 1.1.4.

Proof of Theorem 1.1.4. See [here](#). ■

Bibliography

[New26] I. Newton. *Philosophiae naturalis principia mathematica*. Innys, 1726. URL: <https://books.google.com/books?id=WeZ09rjv-1kC>.